

Lecture 10: Introduction to Actuarial Science

and Insurance

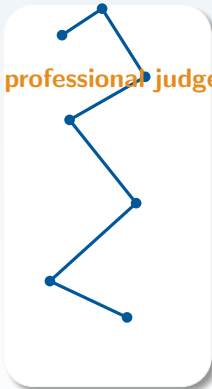
From uncertainty, risk sharing, pooling, expected loss, pricing, and professional judgement

This Phase 3 lecture assumes students have **not heard the term insurance before**. We therefore begin with ordinary uncertainty, build the idea of a possible financial loss, and then show how actuarial science uses data, probability, money, and judgement to design fair and sustainable promises.

Design principles: story-first • micro-steps • no formula memorisation as the main goal • multiple representations • technology checks • ethical discussion.

Author: Department of Statistics and Data Science, The Chinese University of Hong Kong

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Key Definition / Result

What this lecture covers

- ▶ Opening problem and motivation
- ▶ 0. From uncertainty to risk
- ▶ 1. From risk to risk management choices
- ▶ A. What insurance does
- ▶ B. Expected value and pricing
- ▶ C. Pooling and the Law of Large Numbers
- ▶ D. When insurance markets fail
- ▶ E. Time value of money
- ▶ F. Modelling layers and technology
- ▶ Review sets and synthesis

Everyday uncertainty

Possible loss

Risk management choices

Insurance promise

Expected loss

Premium and loading

Pooling and stability

Reality and ethics

Investigation / Activity

Pedagogical features: applications-first, micro-steps, technology checks, contextual interpretation, calm professional visuals, and rich questioning for gifted learners.

Key Definition / Result

Central question: How much should we pay today to manage uncertain future losses fairly and sustainably?

Horizon Bay and the phone-protection plan

A coastal community, **Horizon Bay**, faces tropical storms and occasional earthquakes. At the same time, your class is considering a **phone-screen protection plan**. These look different, but both ask the same financial question.

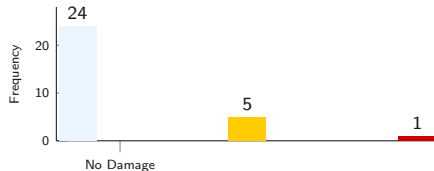
Cognitive Trigger / Opening Problem

How can a group turn *individual uncertainty* into a *manageable financial plan*?

Driving questions

- ▶ How often do losses happen?
- ▶ How large are the losses when they happen?
- ▶ How many people share the risk?
- ▶ What premium would be fair but still safe?

Phone Screen Damage (n=30)



Investigation / Activity

Interpretation: The class scheme is a *small-scale* insurance problem. Horizon Bay is a *community-scale* insurance problem. Actuarial science studies both.

Key Definition / Result

Uncertainty means that more than one future outcome is possible and we do not know which one will occur. Insurance begins only after we ask whether that uncertainty may create a meaningful financial loss.

Investigation / Activity

Not all uncertainty is financial

- ▶ Will it rain tomorrow?
- ▶ Will the bus be late?
- ▶ Will a quiz be difficult?

Investigation / Activity

Insurance focuses on loss uncertainty

- ▶ repair cost
- ▶ medical bill
- ▶ property damage
- ▶ income interruption

Common Mistake / Pitfall

Important distinction: Being unsure is not enough. For insurance, we usually need a possible **financial loss** that can be described and estimated.

Cognitive Trigger / Opening Problem

Student prompt Think of three uncertain events in school life. Which of them could become a financial loss? Which are merely inconvenience or disappointment?

Key Definition / Result

Risk, for this lecture: Risk is uncertainty concerning a possible loss. In actuarial work, we try to describe the financial consequence and estimate how likely it is.

Investigation / Activity

Loss exposure: A loss exposure is a situation in which a loss is possible, even if no loss has happened yet. A building, a phone, a car, a business, or a person can all create different loss exposures.

Technology Check

Simple	translation	table
Everyday phrase	Actuarial language	
"My screen might break"	loss exposure	
"How likely?"	probability / frequency	
"How expensive?"	severity	
"Can we share it?"	pooling / insurance	

Cognitive Trigger / Opening Problem

Deep question: If two students both face a 5% chance of loss, but one loss is HKD 100 and the other is HKD 50,000, are the risks really similar?

Key Definition / Result

Layer 1: event phone falls
fire starts
accident occurs
illness happens

Key Definition / Result

Layer 2: loss repair
replacement
medical bill
income loss

Key Definition / Result

Layer 3: financial plan individual
alone
family
group pool
insurer

Cognitive Trigger / Opening Problem

Key bridge: Insurance begins when people decide that some losses are too large or too unpredictable to face alone, and that a structured group plan may be more reasonable.

Key Definition / Result

Insurance is only one technique for treating risk. Before students learn pricing, they should see that a person or organisation has several possible responses to risk.

Investigation / Activity

Avoidance

Do not take the activity or exposure.

Investigation / Activity

Loss control

Reduce the chance or size of loss.

Investigation / Activity

Retention

Keep the risk and pay losses yourself.

Investigation / Activity

Transfer

Shift part of the financial burden to another party.

Cognitive Trigger / Opening Problem

Coherence bridge: Insurance is a specialised form of risk transfer that often works together with loss control, retention through deductibles, and careful policy design.

Common Mistake / Pitfall

Individual saving can work when losses are small, frequent, and affordable. For example, a student may keep a small emergency fund for stationery, transport, or minor repairs.

Common Mistake / Pitfall

Individual saving becomes difficult when losses are rare but severe. A family may not have enough savings for a house fire, major medical bill, or income interruption.

Key Definition / Result

Bridge to insurance: When many people face similar but not perfectly simultaneous losses, pooling small contributions can create a fund that protects the few who suffer large covered losses.

Key Definition / Result

Actuarial science sits at the intersection of **mathematics**, **finance**, and **real-world decision-making**. The goal is not merely to compute answers. The goal is to make **financially sound decisions under uncertainty**.

In everyday life, uncertainty appears in

- ▶ accidents and illness
- ▶ damaged property
- ▶ transport delays
- ▶ investment outcomes
- ▶ natural disasters

Technology Check

Technology note Throughout this lecture, students can verify ideas using:

- ▶ Microsoft Copilot or Google Gemini
- ▶ spreadsheets
- ▶ R or an online R compiler

Cognitive Trigger / Opening Problem

Gifted learner challenge Does mathematics remove uncertainty? Or does it help us *organise, price, and communicate* uncertainty more intelligently?

Many real-life situations can be modelled as a choice between facing a large uncertain loss alone, or sharing that risk with many others.

Key Definition / Result

Definition of insurance: Insurance is a financial arrangement in which many people contribute to a common pool. The pool is then used to compensate the few who suffer covered losses.

Investigation / Activity

Core mechanism

Many pay small $\Rightarrow$ Few receive large

Key Definition / Result

Risk transfer

The policyholder transfers part of the financial burden to the insurer.

Key Definition / Result

Risk pooling

The insurer combines many similar exposures into one larger group.

Key Definition / Result

Indemnification

A valid claim aims to restore the policyholder's financial position after a covered loss.

Common Mistake / Pitfall

Misconception 1: “If I buy insurance, the risk disappears.”

Correction: The event may still happen. Insurance mainly changes **who pays** and **how the cost is shared**.

Common Mistake / Pitfall

Misconception 2: “If I do not claim, I wasted my premium.”

Correction: The premium bought protection against a possible severe cost. No claim means the bad event did not happen to you this time.

Cognitive Trigger / Opening Problem

Mini debate: If a student's phone is not damaged this year, did the student receive any value from the protection plan? Give a mathematical answer and a practical answer.

Key Definition / Result

Policyholder side

- ▶ pays a premium
- ▶ follows contract conditions
- ▶ reports a covered claim honestly
- ▶ may still bear a deductible or uncovered part

Key Definition / Result

Insurer side

- ▶ collects premiums into a pool
- ▶ estimates future claims
- ▶ pays valid claims
- ▶ holds enough reserves and capital

Cognitive Trigger / Opening Problem

Coherence bridge: The actuarial calculations we study later support the insurer side of the promise: pricing, reserving, solvency, and fairness.

Key Definition / Result

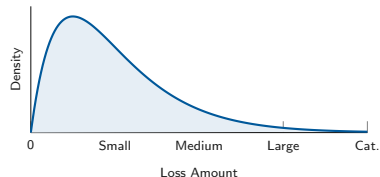
What actuaries do

- ▶ analyse past data
- ▶ estimate how often losses occur (frequency)
- ▶ estimate how large losses are (severity)
- ▶ calculate premiums and reserves
- ▶ advise organisations on uncertainty

Investigation / Activity

Key vocabulary: premium • claim • payout • policy • deductible • loading • reserve • solvency

Modelling Claim Severity



Cognitive Trigger / Opening Problem

Deep question: If the future is unknown, what gives actuaries the confidence to put a price on it?

Worked Example

Example A.1 A class of 30 students wants to set up a phone-screen protection plan. Historical data suggest that in one year:

- ▶ 24 students will have no damage
- ▶ 5 students will need a minor repair of HKD 300
- ▶ 1 student will need a major repair of HKD 1,200

Question What does this tell us about a reasonable annual contribution per student *before* expenses? **Micro-steps**

1. Compute total expected class repairs.
2. $5 \times 300 + 1 \times 1200 = 2700$.
3. Spread that across 30 students.
4. $2700 \div 30 = 90$.

Final answer: HKD 90 per student as the pure expected cost.

Technology Check

Frequency		×	Cost		Calculation
Damage	Freq		Cost	Total	
None	24		0	0	
Minor	5		300	1500	
Major	1		1200	1200	
Sum Total:				2700	

Investigation / Activity

Interpretation: HKD 90 is *not* what any one student will necessarily cost. It is the **long-run average expected cost per student**.

Probability $\Rightarrow$ Expected loss $\Rightarrow$ Premium

Key Definition / Result

Previously in mathematics: Probability was a number between 0 and 1.

Investigation / Activity

Now in actuarial science: Probability becomes money when we attach a financial consequence to each possible outcome.

Cognitive Trigger / Opening Problem

Class discussion Why is a probability of 0.05 very different in practice when the loss is HKD 100 versus HKD 10,000,000?

Key Definition / Result

Expected value represents the long-run average outcome of a random process.

$$E[X] = \sum p_i x_i$$

where p_i is the probability of outcome i , and x_i is the monetary outcome or payout.

Investigation / Activity

Multiple representations

- ▶ table of outcomes and probabilities
- ▶ formula with summation
- ▶ spreadsheet SUMPRODUCT
- ▶ verbal interpretation as a long-run average cost

Common Mistake / Pitfall

Warning Expected value does **not** describe what happens to a single person in one trial. It describes what the system averages to over many similar cases.

Worked Example

Example B.1 Suppose an insurer models three outcomes for a device policy:

- ▶ 80%: no damage, payout = 0
- ▶ 15%: minor damage, payout = 100
- ▶ 5%: major damage, payout = 500

Micro-steps

1. Write the expected value formula.
2. $E[X] = 0.80(0) + 0.15(100) + 0.05(500)$
3. $E[X] = 0 + 15 + 25 = 40$

Final answer: Expected payout = 40

Technology Check

Spreadsheet

SUMPRODUCT

Outcome	Prob.	Payout
No damage	0.80	0
Minor	0.15	100
Major	0.05	500

=SUMPRODUCT(B2:B4, C2:C4)

Investigation / Activity

Interpretation: On average, each policy costs 40 in the long run.

Key Definition / Result

$$\text{Premium} = \text{Expected loss} + \text{Loading}$$

Investigation / Activity

What loading covers

- ▶ uncertainty and safety margin
- ▶ administrative expenses
- ▶ claims handling costs
- ▶ capital costs or profit margin

Common Mistake / Pitfall

Why not charge exactly expected value?

- ▶ actual outcomes vary from year to year
- ▶ data may be incomplete or noisy
- ▶ prices and repair costs may change
- ▶ a bad year can happen even if the average is correct

Cognitive Trigger / Opening Problem

If the expected loss is 40, what could go wrong if the insurer charges exactly 40 to everyone?

Worked Example

Example B.2 A community fire insurance plan covers 1,000 houses. Historical data suggest:

- ▶ expected number of fires each year = 5
- ▶ average loss per fire = 20,000
- ▶ loading = 20% of the pure premium

Micro-steps

1. Total expected loss = $5 \times 20,000 = 100,000$
2. Pure premium per house = $100,000 / 1,000 = 100$
3. Add 20% loading = $100 \times 1.20 = 120$

Final answer: Annual premium per house = 120

Technology Check

Technology path Spreadsheet cells could show:

- ▶ number of claims
- ▶ average severity
- ▶ expected total loss
- ▶ per-unit premium
- ▶ loaded premium

Investigation / Activity

Interpretation: Expected losses are translated into a fair group contribution, then adjusted for safety and sustainability.

Key Definition / Result

Pure premium

The mathematically expected claim cost per exposure.

$$\frac{\text{Expected total loss}}{\text{Number of exposures}}$$

Key Definition / Result

Gross premium

The amount actually charged after expenses, safety margin, and other business considerations.

$$\text{Pure premium} + \text{loading}$$

Common Mistake / Pitfall

Promise quality: A lower premium is not automatically better if it makes the promise unreliable. Insurance must be affordable *and* financially sustainable.

Cognitive Trigger / Opening Problem

Gifted learner challenge Design three premiums for the same risk: extremely cheap, mathematically fair, and professionally responsible. Explain the trade-off.

Key Definition / Result

Ordinary product pricing: A seller often knows many production costs before selling the item. The price can be built from known costs plus margin.

Key Definition / Result

Insurance pricing: The insurer sells the promise before knowing the exact future claims. The price is therefore built from estimated probabilities, estimated severities, expenses, safety margins, and capital needs.

Cognitive Trigger / Opening Problem

Coherence bridge: This is why actuarial science needs both mathematics and judgement: the number is calculated before the full cost is known.

Key Definition / Result

The Law of Large Numbers (LLN): As the number of observations or exposure units becomes large, the average outcome tends to get closer to the expected value.

Investigation / Activity

Intuition

- ▶ 10 coin flips: quite erratic
- ▶ 10,000 coin flips: usually close to 50% heads

Insurance works in the same spirit.

Key Definition / Result

Big idea: Insurance does **not** eliminate risk. It re-distributes and stabilises risk by making the *average loss per person* more predictable.

Cognitive Trigger / Opening Problem

What gets more predictable as the pool grows: the total loss, the average loss per person, or both?

Worked Example

Example C.1 For one driver:

- ▶ 10% chance of a major incident costing 10,000
- ▶ 90% chance of no incident costing 0

Micro-steps

1. Expected loss for one driver: $E[X] = 0.1(10,000) + 0.9(0) = 1,000$
2. Variance for one driver: $E[X^2] = 0.1(10,000^2) = 10,000,000$, so $\text{Var}(X) = 10,000,000 - (1,000)^2 = 9,000,000$
3. Standard deviation for one driver: $SD(X) = 3,000$
4. For 100 independent drivers: average loss per policy still has mean 1,000, but $SD(\bar{X}) = 300$

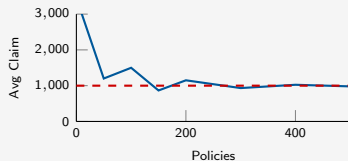
Final answer: Expected loss per policy stays the same, but the portfolio average becomes much less variable.

Technology Check

LLN

Simulation

Stabilising at 1,000



Investigation / Activity

Key insight

Total uncertainty grows, but **relative uncertainty** shrinks.

Investigation / Activity

A helpful pool usually has

1. many exposure units
2. losses that are not all triggered together
3. measurable and reasonably estimable costs
4. a premium that people can actually afford

Common Mistake / Pitfall

Why correlation matters: If a single event can damage everyone at once, pooling is weaker. The average may no longer be stable enough for ordinary insurance pricing.

Cognitive Trigger / Opening Problem

Example prompt: Is a typhoon risk more like independent phone-screen damage or more like a common shock to the whole community?

Individual claims: spread out



Average of many claims: tightly clustered



Investigation / Activity

What the graph shows

- ▶ many simulated small samples: very spread out
- ▶ larger sample averages: tightly clustered near the mean

Key Definition / Result

Teacher talk track: “A single person’s experience is noisy. A large portfolio’s average experience is much more stable. That is the statistical foundation of insurance.”

Cognitive Trigger / Opening Problem

If a risk is *highly correlated* across policyholders, does pooling still help as much?

Investigation / Activity

Use technology to explore: Simulate claims for a portfolio where each policy has a 20% claim probability and claim severities are 1,000, 5,000, or 10,000.

Key Definition / Result

Task ideas

- ▶ simulate 10 policies for 100 years
- ▶ compute average total claims
- ▶ repeat with 100 policies
- ▶ compare the histograms

Technology Check

Suggested prompt: “Simulate 100 years of claims for 10 policies, each with claim probability 0.2, and claim severities 1000, 5000, 10000 with probabilities 0.7, 0.2, 0.1. Show a histogram and the average total claim.”

Cognitive Trigger / Opening Problem

Discussion prompt Does the average simulated total claim move closer to the theoretical expected value when the portfolio becomes larger?

Key Definition / Result

All actuarial models rely on assumptions

- ▶ probabilities are estimated reasonably well
- ▶ claim events are not too strongly dependent
- ▶ costs are measurable
- ▶ large catastrophic correlations are manageable

Investigation / Activity

Six desirable insurability features

1. many exposure units
2. accidental loss
3. measurable loss
4. limited catastrophe concentration
5. calculable chance of loss
6. economically feasible premium

Common Mistake / Pitfall

What can go wrong?

- ▶ data errors
- ▶ incorrect assumptions
- ▶ changing behaviour
- ▶ inflation or repair-cost spikes
- ▶ correlated disasters

Worked Example

Example D.1 Assess whether each of the following is generally insurable.

1. Damage to one house from a local hailstorm
2. A major global stock market crash next week
3. The “risk” of not finding a rare ancient artifact in your garden

Micro-steps

- ▶ Hail damage: many exposure units, measurable loss, usually insurable.
- ▶ Global crash: highly correlated, catastrophic, hard to pool using ordinary insurance.
- ▶ Rare artifact: not a clear measurable loss with a calculable chance in the usual insurance sense.

Final answer: (1) generally insurable; (2) not usually insurable by traditional insurance; (3) not insurable.

Investigation / Activity

Critical idea: Not every uncertain event is suitable for insurance. The mathematics of pooling only works well when the underlying structure is suitable.

Cognitive Trigger / Opening Problem

Which insurability criterion is most challenged by cyberattacks or climate-related disasters?

Key Definition / Result

Adverse selection: Higher-risk individuals are more likely to seek insurance, while lower-risk individuals may choose not to buy it. The pool becomes riskier than expected.

Key Definition / Result

Moral hazard: After obtaining insurance, behaviour may change in ways that increase the chance or size of loss.

Common Mistake / Pitfall

Adverse selection example: If a health policy is priced for an “average” person, but mostly high-risk people buy it, the insurer’s claim cost rises unexpectedly.

Common Mistake / Pitfall

Moral hazard example: If a person is heavily insured, they may take fewer precautions, increasing the expected loss after the contract starts.

Investigation / Activity

Mitigation tools: underwriting • deductibles • coinsurance • exclusions • risk-based pricing • monitoring and prevention

Worked Example

Example D.2 A health insurer charges everyone HKD 300 per month. But there are two groups:

- ▶ low-risk individuals: expected claims = 100 per month
- ▶ high-risk individuals: expected claims = 400 per month

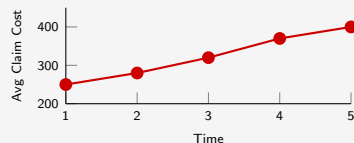
Micro-steps

1. Compare the premium with each group's expected cost.
2. Low-risk people may feel 300 is too expensive and opt out.
3. High-risk people find 300 attractive and stay in.
4. The insurer's average claim cost rises above what it assumed.
5. Premiums may then rise even further, pushing out more low-risk people.

Final answer: A poorly priced market can spiral into an unstable high-risk pool.

Technology Check

Adverse Selection Spiral
Average risk increasing



Investigation / Activity

Discussion: If everyone pays the same, is that fair? If everyone pays according to personal risk, is that fair?

Key Definition / Result

Coverage

What losses are protected?

Key Definition / Result

Exclusions

What losses are not protected?

Key Definition / Result

Deductible

What does the policyholder pay first?

Key Definition / Result

Limit

What is the maximum amount paid?

Common Mistake / Pitfall

Why boundaries matter: Without boundaries, the premium would be unclear, behaviour could change, and the promise may become financially unsustainable.

Cognitive Trigger / Opening Problem

Class design task For the phone-screen plan, propose one exclusion, one deductible, and one coverage limit. Explain how each changes fairness and cost.

Key Definition / Result

Policy design changes the random variable that actuaries price. A deductible, a limit, or an exclusion does not merely change wording; it changes the expected payout, the variability of losses, and the behaviour of policyholders.

Investigation / Activity

Deductible

transfers small losses back to the policyholder.

Investigation / Activity

Limit

caps extreme payouts and protects solvency.

Investigation / Activity

Exclusion

removes losses that are too hard, unfair, or expensive to pool.

Cognitive Trigger / Opening Problem

Coherence bridge: Contract wording and actuarial modelling are connected. A mathematical price only makes sense after defining what is covered.

Key Definition / Result

Time value of money: Money today is worth more than the same amount in the future because it can be invested.

$$PV = \frac{F}{(1+i)^n}$$

Investigation / Activity

Why TVM matters in insurance

- ▶ premiums are often collected today
- ▶ claims may be paid years later
- ▶ reserves must be held now for later obligations

Cognitive Trigger / Opening Problem

Useful metaphor: Insurance pricing is a bit like a **time machine for money**: we move future cash flows to a common date so we can compare them fairly.

Worked Example

Example E.1 An insurer expects to pay a claim of 50,000 in 3 years. The annual interest rate is 4%. **Micro-steps**

1. Identify variables. $F = 50,000$, $i = 0.04$, $n = 3$
2. Apply formula. $PV = 50,000(1.04)^{-3}$
3. Evaluate discount factor. $(1.04)^{-3} \approx 0.888996$
4. Multiply. $PV \approx 50,000 \times 0.888996 = 44,449.80$

Final answer: The insurer needs about **44,449.80** today.

Technology Check

Technology path

Spreadsheet:

=50000/(1.04)^3 R:

50000*(1.04)^(-3)

Investigation / Activity

Interpretation: A larger interest rate means the insurer can set aside less money today for the same future payment.

Worked Example

Example E.2 A 1-year term life policy pays 100,000 if death occurs within the year. Probability of death = 0.001. Interest rate = 3%. **Micro-steps**

1. Expected death benefit before discounting: $100,000 \times 0.001 = 100$
2. Discount to present value: $NSP = 100(1.03)^{-1}$
3. Evaluate: $(1.03)^{-1} \approx 0.970874$, so $NSP \approx 97.09$

Final answer: Net single premium ≈ 97.09

Investigation / Activity

Integrated insight: This example combines probability + expected value + time value of money.

Cognitive Trigger / Opening Problem

What extra amount would an insurer add in practice beyond the net single premium?

Key Definition / Result

Reserve intuition: If an insurer promises to pay future claims, it should not spend all collected premiums immediately. Part of the money must be set aside for future obligations.

Investigation / Activity

Actuarial connection: Reserves depend on expected future cash flows, timing, interest rates, uncertainty, and the exact contract design.

Cognitive Trigger / Opening Problem

Coherence bridge: Premium answers “what should be collected?” Reserve answers “what should be held now so the promise remains credible?”

Key Definition / Result

$$\text{Total loss} \approx \text{Frequency} \times \text{Severity}$$

Investigation / Activity

Frequency model: How many claims occur? Often modelled with discrete random variables such as Bernoulli or Binomial distributions.

Investigation / Activity

Severity model: How large is each claim? Can be modelled using a discrete payout table for beginners.

Cognitive Trigger / Opening Problem

Once both layers are modelled, we can estimate total expected losses, simulate many possible futures, and study extreme outcomes.

Worked Example

Example F.1 A portfolio has 10 independent car policies. Each policy has a 0.20 probability of a claim. **Questions**

- ▶ What is the expected number of claims?
- ▶ What is the probability of exactly 3 claims?

Micro-steps

1. Recognise a Binomial model: $X \sim B(n = 10, p = 0.20)$
2. Expected claims: $E[X] = np = 10(0.20) = 2$
3. Probability of exactly 3 claims: $P(X = 3) = \binom{10}{3}(0.20)^3(0.80)^7$
4. $\binom{10}{3} = 120$, so $P(X = 3) \approx 0.2013$

Final answer: Expected claims = 2; $P(X = 3) \approx 0.2013$

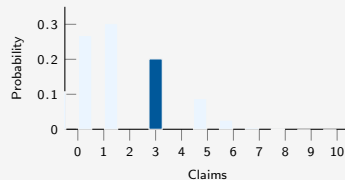
Technology Check

Binomial

PMF:

$B(10, 0.2)$

Exactly 3 claims



Investigation / Activity

Frequency tells us *how many* claim events we expect, before thinking about claim size.

Worked Example

Example F.2 Suppose one claim amount S has the following distribution:

- ▶ $P(S = 1000) = 0.7$
- ▶ $P(S = 5000) = 0.2$
- ▶ $P(S = 10000) = 0.1$

Micro-steps

1. Write the expected value expression.
2. $E[S] = 1000(0.7) + 5000(0.2) + 10000(0.1)$
3. $E[S] = 700 + 1000 + 1000 = 2700$

Final answer: Expected severity of one claim = 2,700

Investigation / Activity

Aggregate expected loss: If expected frequency is 2 claims and expected severity per claim is 2,700, then

Expected total claims $\approx 2 \times 2700 = 5400$.

Cognitive Trigger / Opening Problem

Would two portfolios with the same expected total claims necessarily carry the same risk? Why or why not?

Technology Check

Useful Copilot / Gemini prompts

- ▶ Explain expected value in a school insurance context.
- ▶ Create a table and chart for a 3-outcome claim distribution.
- ▶ Simulate 100 years of claims for a small portfolio.
- ▶ Compare loaded premium and pure premium.
- ▶ Explain adverse selection using a simple story.

Technology Check

Useful spreadsheet / R ideas

- ▶ SUMPRODUCT for expected value
- ▶ charts for frequency and severity distributions
- ▶ present value formula tables
- ▶ simulation using random numbers
- ▶ comparison between small and large pools

Common Mistake / Pitfall

Teacher note: Replace placeholders with actual screenshots from the tool the class will use. The purpose of technology is to support interpretation, not to replace it.

Key Definition / Result

Why ethics matters: Actuaries work on problems that affect financial security, fairness, and long-term trust. Technical correctness is necessary, but not sufficient.

Investigation / Activity

Professional values

- ▶ integrity
- ▶ competence
- ▶ objectivity
- ▶ confidentiality
- ▶ clear communication
- ▶ public interest

Cognitive Trigger / Opening Problem

Discussion prompt: If more personal data leads to more accurate pricing but also makes insurance unaffordable for vulnerable groups, what should a responsible professional do?

Key Definition / Result

Questions

1. Define insurance in one clear sentence.
2. Explain the difference between a premium and a claim.
3. Calculate $E[X]$ if outcomes are 0 (0.7), 200 (0.2), 500 (0.1).
4. Why is expected value a long-run idea rather than a one-person prediction?
5. A policy has expected loss 80 and loading 15. Find the premium.
6. State the Law of Large Numbers in simple words.
7. Why does a large portfolio make average losses more predictable?
8. Write one sentence explaining why variance matters in insurance.

Investigation / Activity

Quick strategy

- ▶ formula
- ▶ calculation
- ▶ context sentence

Technology Check

Teacher suggestion

Do Q3 and Q5 both by hand and in a spreadsheet.

Key Definition / Result

Questions

9. A claim of 12,000 is due in 2 years at 5%. Find its present value.
10. A portfolio has 20 policies with claim probability 0.1 each. What is the expected number of claims?
11. Explain why “many exposure units” helps insurability.
12. Give one example of moral hazard.
13. Give one example of adverse selection.
14. Briefly explain why a global stock market crash is difficult to insure traditionally.
15. If expected frequency is 3 and expected severity is 600, estimate the expected total claims.
16. In one sentence, explain the role of loading.

Common Mistake / Pitfall

Common habits to avoid

- ▶ writing numbers without units
- ▶ giving formulas without interpretation
- ▶ mixing up expected value and probability

Investigation / Activity

Questions

1. Horizon Bay has 5,000 houses. If 2% experience storm damage each year and average damage is 15,000, find the expected annual claims.
2. If a 15% loading is added, what premium per house is needed?
3. Why is Horizon Bay's self-insurance plan more feasible than an individual homeowner acting alone?
4. A 2-year life policy pays 200,000. Probability of death in Year 1 is 0.002; in Year 2 given survival is 0.0025. Interest rate is 4%. Describe the steps for calculating the net single premium.
5. Explain how deductibles reduce moral hazard.
6. Propose one underwriting question that might help reduce adverse selection in motor insurance.
7. Explain why two portfolios with the same expected loss may still require different pricing.

Cognitive Trigger / Opening Problem

Stretch prompt: Where exactly does fairness appear in these calculations: in probabilities, in premiums, in policy design, or in regulation?

Investigation / Activity

Questions

8. A portfolio of 5 policies has claim probability 0.3 for each policy. Find the probability of exactly 2 claims.
9. A risk has low frequency but very high severity. What challenges does this create for pricing?
10. Explain why cyber risk may challenge the criterion of “limited catastrophe concentration.”
11. If interest rates rise, what happens to the present value of a future claim? Explain.
12. Suggest a simple simulation plan for modelling a rare equipment insurance product with limited historical data.
13. Why should actuaries communicate uncertainty clearly instead of presenting one number as certain?
14. Design an insurance product for one school-related risk and describe: the pool, the premium idea, one exclusion, and one anti-moral-hazard feature.

Technology Check

Extension: Turn Q8 or Q14 into a spreadsheet mini-project or a short Copilot-assisted investigation.

Key Definition / Result

Return to the opening problem: Horizon Bay and the phone-screen plan are not just stories. They are examples of how actuarial thinking turns uncertain losses into structured financial promises.

Investigation / Activity

Three core takeaways

1. Insurance pools risk.
2. Expected value drives the first layer of pricing.
3. Large numbers create stability, but real life adds complications.

Investigation / Activity

Second-layer takeaways

1. Loading protects sustainability.
2. Time value matters because claims arrive later.
3. Behaviour and information can distort markets.

Cognitive Trigger / Opening Problem

Final reflective prompt: Actuarial science is not mainly about *getting a number*. It is about **making a defensible decision under uncertainty**. What makes a decision defensible?

Key Definition / Result

Fast recap

- ▶ What is uncertainty?
- ▶ What is a loss exposure?
- ▶ What is insurance?
- ▶ What is an expected payout?
- ▶ Why do we add loading?
- ▶ Why does pool size matter?
- ▶ Why does present value appear in insurance work?

Investigation / Activity

Possible next chapter links

- ▶ deeper modelling and simulation
- ▶ survival probabilities and life contingencies
- ▶ Value-at-Risk and solvency
- ▶ professional risk management frameworks

Thank you

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Key Definition / Result

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